Section 6.7 — Piecewise Functions — Answer Key

Math 103: College Algebra

Building a Piecewise Function

For each list of definitions below, combine the three parts into a single piecewise function. Then, state the domain of the piecewise function.

1. $f(x) = \begin{cases} 2x-4 & \text{if} \quad 2 < x < 4 \\ x^2-1 & \text{if} \quad 4 \le x < 6 \\ x-2 & \text{if} \quad 6 \le x < 8 \end{cases}$
   $\operatorname{Dom} f = (2, 8)$

2. $g(x) = \begin{cases} 2x-3 & \text{if} \quad x < 2 \\ 5x-2 & \text{if} \quad x > 4 \end{cases}$
   $\operatorname{Dom} g = (-\infty, 2) \cup (5, \infty)$

3. $h(x) = \begin{cases} x & \text{if} \quad x \neq 5 \\ -x & \text{if} \quad x = 5 \end{cases}$
   $\operatorname{Dom} h = \mathbb{R}$

Evaluating a Piecewise Function

Evaluate each function below for the given values. Write "undefined" if the given value is not in the domain.

4. a. $f(-7) = -6$ b. $f(-3) = -6$ c. $f(2)$ is undefined d. $f(0) = 3$

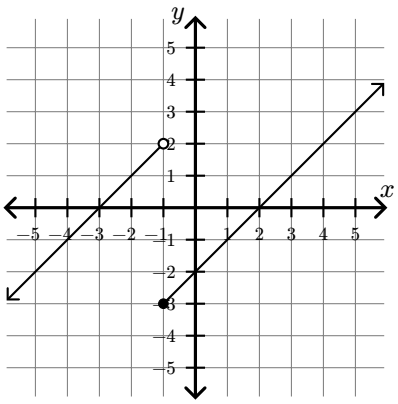
5. a. $g(-2) = 2$ b. $g(12) = 23$ c. $g(4)$ is undefined d. $g(5)$ is undefined

6. a. $h(-1.4) = -1$ b. $h(0.5) = -1$ c. $h(2) = 2$ d. $h(2.5)$ is undefined

Graphing a Piecewise Linear Function

Graph each function below on a seperate sheet of graph paper or on the other side of this page.

7.



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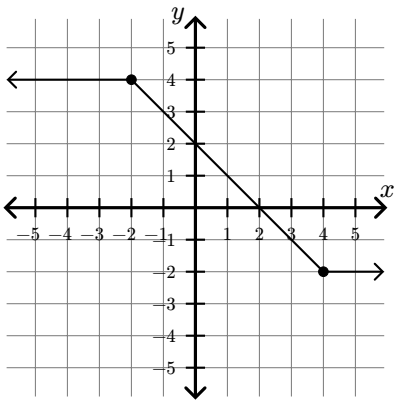

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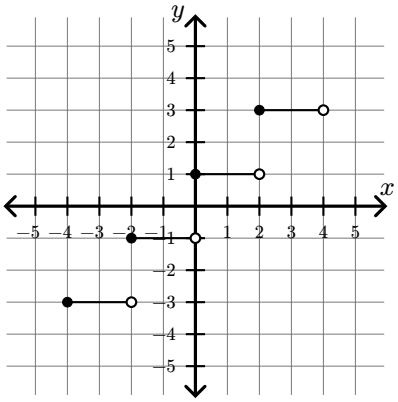

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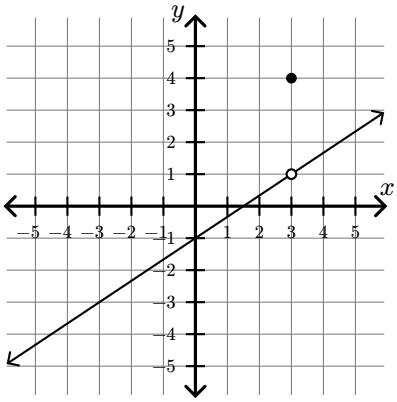

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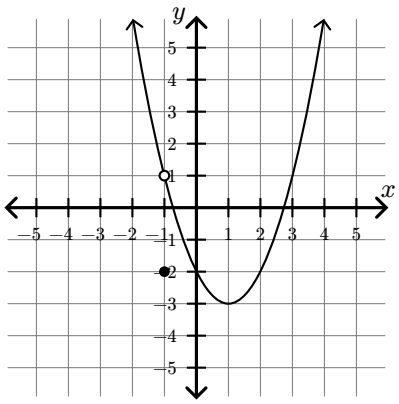
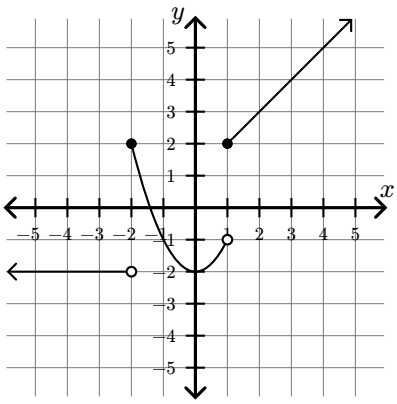
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